

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 7: Ratio and Proportion

Topic-wise short notes with solved examples

A ratio compares two quantities and a proportion says that two ratios are equal. This chapter covers the special kinds of ratio, continued proportion and the properties of proportion used to solve equations.

1 Ratio

- The **ratio** of a to b is $\frac{a}{b}$, written $a : b$, where $b \neq 0$. a is the **antecedent** and b the **consequent**.
- A ratio has no unit. Both quantities must be of the same kind and in the same unit.
- A ratio is unchanged when both terms are multiplied or divided by the same non-zero number.

Ratio	For $a : b$ it is
Duplicate ratio	$a^2 : b^2$
Triplicate ratio	$a^3 : b^3$
Sub-duplicate ratio	$\sqrt{a} : \sqrt{b}$
Sub-triplicate ratio	$\sqrt[3]{a} : \sqrt[3]{b}$
Reciprocal ratio	$b : a$

- The **compound ratio** of $a : b$ and $c : d$ is $ac : bd$.

Example 1.1 If $a : b = 3 : 5$, find $(5a + 2b) : (3a + 4b)$.

SOLUTION

Let $a = 3k$ and $b = 5k$.

$$\frac{5a + 2b}{3a + 4b} = \frac{15k + 10k}{9k + 20k} = \frac{25k}{29k}$$

25 : 29

Example 1.2 Find the duplicate ratio of $2 : 3$ and the compound ratio of $2 : 3$ and $9 : 4$.

SOLUTION

$$\text{Duplicate ratio} = 2^2 : 3^2 = 4 : 9$$

$$\text{Compound ratio} = (2 \times 9) : (3 \times 4) = 18 : 12$$

4 : 9 and 3 : 2

2 Proportion

- Four quantities a, b, c, d are in **proportion** if $a : b = c : d$, written $a : b :: c : d$.
- a and d are the **extremes** and b and c the **means**.

$$\text{Product of extremes} = \text{product of means} : \quad ad = bc$$

- d is the **fourth proportional** to a, b and c .

Example 2.1 Find the fourth proportional to 3, 5 and 12.

SOLUTION

Let it be x . Then $3 : 5 = 12 : x$, so $3x = 60$.

$$x = 20$$

Example 2.2 What number must be added to each of 2, 4, 6 and 10 so that the results are in proportion?

SOLUTION

Let the number be x . Then $(2 + x)(10 + x) = (4 + x)(6 + x)$.

$$20 + 12x + x^2 = 24 + 10x + x^2, \text{ so } 2x = 4.$$

The number is 2. (Check: 4, 6, 8, 12 and $4 \times 12 = 6 \times 8$.)

3 Continued proportion

- a, b, c are in **continued proportion** if $a : b = b : c$.

$$b^2 = ac$$

- b is the **mean proportional** between a and c : $b = \sqrt{ac}$.
- c is the **third proportional** to a and b : $c = \frac{b^2}{a}$.

Example 3.1 Find (i) the mean proportional between 4 and 9 (ii) the third proportional to 4 and 6.

SOLUTION

$$(i) \sqrt{4 \times 9} = \sqrt{36} = 6$$

$$(ii) \frac{6^2}{4} = 9$$

(i) 6 (ii) 9

Example 3.2 If b is the mean proportional between a and c , prove that $\frac{a^2 + b^2}{b^2 + c^2} = \frac{a}{c}$.

SOLUTION

$$b^2 = ac$$

$$\begin{aligned} \text{LHS} &= \frac{a^2 + ac}{ac + c^2} = \frac{a(a + c)}{c(a + c)} \\ &= \frac{a}{c} = \text{RHS} \end{aligned}$$

4 Properties of proportion

- If $\frac{a}{b} = \frac{c}{d}$, then each of the following is true.

Property	Result
Invertendo	$\frac{b}{a} = \frac{d}{c}$
Alternendo	$\frac{a}{c} = \frac{b}{d}$
Componendo	$\frac{a + b}{b} = \frac{c + d}{d}$
Dividendo	$\frac{a - b}{b} = \frac{c - d}{d}$
Componendo and dividendo	$\frac{a + b}{a - b} = \frac{c + d}{c - d}$

Example 4.1 If $\frac{3x + 4y}{3x - 4y} = \frac{7}{3}$, find $x : y$.

SOLUTION

Apply componendo and dividendo: $\frac{(3x + 4y) + (3x - 4y)}{(3x + 4y) - (3x - 4y)} = \frac{7 + 3}{7 - 3}$

$$\frac{6x}{8y} = \frac{10}{4}, \text{ so } \frac{x}{y} = \frac{10}{4} \times \frac{8}{6} = \frac{10}{3}$$

$$x : y = 10 : 3$$

Example 4.2 Solve $\frac{\sqrt{x+5} + \sqrt{x-16}}{\sqrt{x+5} - \sqrt{x-16}} = \frac{7}{3}$.

SOLUTION

Apply componendo and dividendo: $\frac{2\sqrt{x+5}}{2\sqrt{x-16}} = \frac{7+3}{7-3} = \frac{5}{2}$

Squaring: $\frac{x+5}{x-16} = \frac{25}{4}$, so $4x + 20 = 25x - 400$ and $21x = 420$

$$x = 20$$

REMEMBER

Componendo and dividendo is the quickest method whenever one side of an equation has the form $\frac{p+q}{p-q}$.

5 The k-method for proofs

- If $\frac{a}{b} = \frac{c}{d}$, put each ratio equal to k . Then $a = bk$ and $c = dk$. Substitute in both sides of the result to be proved.
- For continued proportion $\frac{a}{b} = \frac{b}{c} = k$: $b = ck$ and $a = bk = ck^2$.

Example 5.1 If $\frac{a}{b} = \frac{c}{d}$, prove that $\frac{a^2 + c^2}{b^2 + d^2} = \frac{ac}{bd}$.

SOLUTION

Let $\frac{a}{b} = \frac{c}{d} = k$, so $a = bk$ and $c = dk$.

$$\text{LHS} = \frac{b^2k^2 + d^2k^2}{b^2 + d^2} = k^2$$

$$\text{RHS} = \frac{bk \times dk}{bd} = k^2$$

LHS = RHS

6 Common mistakes

- Writing a ratio of quantities in different units without converting.
- Confusing duplicate ($a^2 : b^2$) with sub-duplicate ($\sqrt{a} : \sqrt{b}$).
- Applying componendo to one side only. Every property is applied to both sides.
- Forgetting to check the answer in an equation with square roots.